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Review

Aerobiological modeling I: A review of predictive models

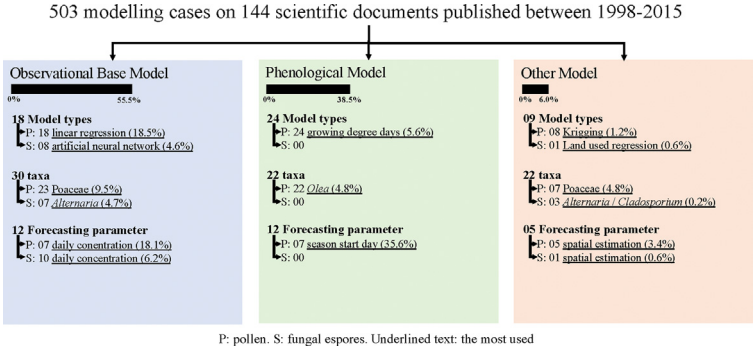
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HIGHLIGHTS

- Predictive models in aerobiology can be classified: Observational Base Models OBM, Phenological Models PHM, Other Models OTM.
- OBM are used with high frequency to forecast concentration and PHM to characterize the main pollen/spore season.
- OBM tend to use a computational statistical model that does not require compliance with normality and linearity in the data.
- PHM are easy to use but they are less frequently used than OBM due to their variety of starting criteria.

GRAPHICAL ABSTRACT



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ABSTRACT

The present work is the first of two reviews on applied modeling in the field of aerobiology. The aerobiological predictive models for pollen and fungal spores, usually defined as predictive statistical models, will, amongst other objectives, forecast airborne particles' concentration or dynamical behavior of the particles. These models can be classified into Observation Based Models (OBM), Phenological Based Models (PHM), or OTher Models (OTM). The aim of this review is to show, analyze and discuss the different predictive models used in pollen and spore aerobiological studies. The analysis was performed on published electronic scientific articles from 1998 to 2016 related to the type of model, the taxa and the modelled parameters. From a total of 503 studies, 55.5% used OBM (44.8% on pollen and 10.7% on fungal spores), 38.5% PHM (all on pollen) and 6% OTM (5.4% on pollen and 0.6% on fungal spores). OBM have been used with high frequency to forecast concentration. The most frequent model of OBM was linear regression (18.5% out of 503) on pollen and artificial neural networks (4.6%) on fungal spores. In the PHM, the principal use was to characterize the main pollen season (flowering season) based on the model of growth degree days. Finally, OTM have been used to estimate concentrations at unmonitored areas. Olea (14,5%) on pollen and Alternaria (4,8%) on fungal spores were the taxa most frequently modelled. Daily concentration was the most modelled parameter by OBM (25.2%) and season start day by PHM (35.6%). The PHM approaches include greater model diversity and use fewer independent variables than OBM. In addition, PHM show to be easier to apply than OBM; however, the wide range of criteria to define the parameters to use in PHM (e.g.: pollination start day) makes that each model is used with a lesser frequency than other models.

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1. Introduction

A wide spectrum of microorganisms and biological particulate matter use the atmosphere as a means of transportation (Cox and Wathes, 1995). Aerobiology studies the presence and behavior of pollen and fungal spores in the air, applying different methods and involving biologists, agronomists, meteorologists and ecologists amongst others (Scheifinger et al., 2013). The aim of the studies in this field is to analyze biological particles (mainly pollen and fungal spores) that are passively transported by the air (Cecchi, 2013). Aerobiological studies have evaluated airborne particle transport and dispersal in several countries over many years. The studies aimed to show the biological characteristics of the airborne particles (Goldstein, 1960; Kanchan and Jayachandra, 1980) and demonstrated that airborne pollen depended on vegetation and environmental conditions, varying between countries, within regions, and even amongst cities within a region (Brown and Hovmøller, 2002; Ribeiro et al., 2003; Prospero et al., 2005; Oliveira et al., 2009b; Ziello et al., 2012; De Linares et al., 2017). At present aerobiological studies are focused on: (i) Understanding the relationship between airborne pollen and fungal spores over the course of a year with the vegetation and with the meteorological parameters over the course of a year (ii) Establishing the relationship and the prevalence of allergic symptoms in humans or of crop pathologies (D'Amato et al., 1998; Tao et al., 2009; Rodríguez-Rajo et al., 2010b); (iii) Predicting yields in agricultural production (Galán et al., 2008); (iv) Carrying out integrated pest management programs on crops (Pasken and Pietrowicz, 2005; Decognet et al., 2009); (v) Developing predictive and dispersion models (Emberlin et al., 2000; Sofiev et al., 2013); (vi) Studying the possible effects of climate change on phenological processes (Aguilera et al., 2013).

Pollen is known to play an important role in respiratory allergies (Bousquet et al., 2007; D'Amato et al., 2007), that appear especially during the flowering periods of plants (Türe and Böcük, 2009). The occurrence of these allergies has also been associated with fungal spores (Hasnain et al., 2012). The studies on prevalence of pollen allergy estimate that it affects up to 40% of the population in Europe (D'Amato et al., 2007). Similar studies on fungal spores are less frequent and in the industrialized world, they are estimated to affect around 25% of the population with atopic allergies (Salvaggio and Aukrust, 1981; Tariq et al., 1996; Kurup et al., 2002).

In addition, Aerobiology can be useful in forecasting crops. The annual airborne pollen is often used as a bioindicator of fruit production. These close relationships have been widely studied, particularly in anemophilous crops. Optimal results have been obtained for European hazelnut (*Corylus avellana* – Lletjos et al., 1993), grape (*Vitis vinifera* – Cunha et al., 2003), olive (*Olea europaea* – Fornaciari et al., 2002, 2005; Galán et al., 2004; Ribeiro et al., 2007a; García-Mozo et al., 2009; Orlandi et al., 2010) and holm-oak (*Quercus ilex* subsp. *ballota* – García-Mozo et al., 2007). In the case of fungal spores, the aerobiological studies have been used to predict the risk of crop infection, e.g. apple scab (*Venturia inaequalis* – Aylor, 1999), tobacco blue mold (*Peronospora tabacina* – Aylor, 1999), grey-mold epidemics (*Botrytis cinerea* – Blanco et al., 2006; Oliveira et al., 2009a; Rodríguez-Rajo et al., 2010b; Fernández-González et al., 2011; Leyronas and Nicot, 2013), soybean rust (*Phakopsora pachyrhizi* – Tao et al., 2009), mildew of vines (*Plasmopara viticola* – Fernández-González et al., 2011) and powdery mildew in vines (*Uncinula necator* – Fernández-González et al., 2011).

Finally, several studies are focused on observing the effect of climate change on the vegetation. Some of them have shown the incidence of temperature increase on airborne pollen records (Spieksma et al., 1995; Frei, 1998; Teranishi et al., 2000; Ziska and Caulfield, 2000; Rasmussen, 2002; Ziello et al., 2012; Fernández-Llamazares et al., 2014), while others have analyzed the incidence on the start of the pollen season, reporting an earlier onset (Emberlin, 1994; Emberlin et al., 1997, 2002; D'Amato et al., 2002; Fitter and Fitter, 2002; Frenguelli, 2002; Rasmussen, 2002; Galán et al., 2005; Breton et al., 2006; García-Mozo et al., 2006; Stach et al., 2007). A meta-analysis developed by Root et al. (2003) found that the average shift (advance) in spring phenology (timing) of blooming in plants of the temperate zone was five days per decade. Other studies centered on Europe in different time periods (from 1991 to 2015) analyzed the trend of fungal spores over years concluding that the increasing temperature trends have an effect on the airborne fungal spore concentrations (Corden and Millington, 2001; Corden et al., 2003; Millington and Corden, 2005; Damialis et al., 2015a, 2015b; Grinn-Gofroń et al., 2011; Sadyś et al., 2016a; Vélez-Pereira et al., 2016).

The present work corresponds to the first of two reviews on applied modeling in the field of aerobiology, in this case aiming at reviewing the predictive models used in the aerobiological studies of pollen and fungal spores. In relation to the pollen prediction models, they are classified, following the criteria by Scheifinger et al. (2013), in "Observation-Based Models (OBM)" and "process-based PHenological Models (PHM)". We have included a third classification category, "OTher Models (OTM)", for those prediction models that do not fulfill these criteria. In the case of airborne fungal spores, we classified them into OBM and OTM. Papers related to statistics for comparing different parameters or for calculating trends are not included.

The bibliography considered has been obtained via databases and journals and corresponds to papers published from 1998 to February of 2016. The start date for the search was set in 1998 due to the availability of electronic journals.

2. Prediction models on airborne pollen and fungal spores

Fig. 1 shows how the number of studies on airborne pollen and fungal spores has been notably increased in the last 20 years, especially those related to modeling. Although aerobiology is a recent discipline, as shown in Fig. 1, the availability of extensive databases makes this increase in publications possible and demonstrates the importance of these multidisciplinary studies. Tables 1 to 3 show all the papers collected and analyzed in this review.

In the tables, the information is presented according to the alphabetical order of the model's name, the type of bioaerosol modelled (taxon: pollen or fungal spore) and the parameter modelled (i.e. daily concentration, date of the peak concentration, peak concentration, amongst others).

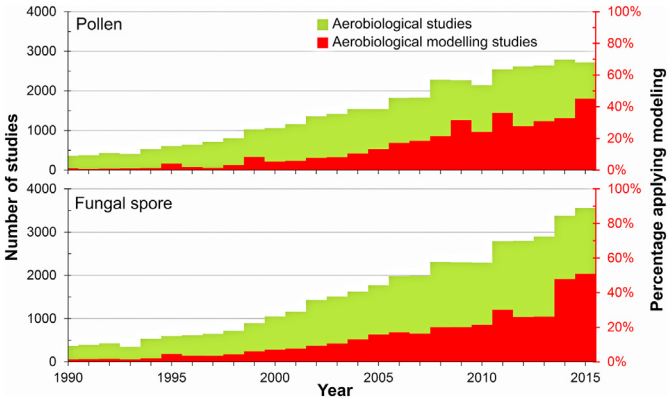


Fig. 1. Number of aerobiological pollen and fungal spore studies (left axis) and the corresponding percentage that apply modeling (right axis) per year reported by Scholar Google. Consultation made on July 22, 2016.

2.1. Observation-Based Models (OBM)

OBM are set as the ratio of biological aerosol concentrations and a set of independent variables that are generally the meteorological parameters (Aboulaich et al., 2013). These models are expressed or described by a function or equation, which allows a prediction from the independent variables. Raynor and Hayes (1970), Davies and Smith (1973) and Bringfelt (1979) were the first articles on bioaerosols modeling employing OBM. They showed the interest and need for developing methods to predict the daily pollen concentrations.

Table 1 presents 18 different OBM found in this review. Linear regression and its variations is the most applied classical model. This is justified because during the first years of aerobiological studies the other more advanced types of analysis were still not known. However, one of the problems of this method is that aerobiological data used do not assume linearity and normality (Grinn-Gofroń and Strzelczak, 2008a, 2008b; Astray et al., 2010). This leads to errors in models without previous data treatment (Galán et al., 2001a; Díaz de la Guardia et al., 2003; Grinn-Gofroń and Strzelczak, 2008a, 2008b). However, this can be solved by transforming the data (mainly daily data) prior to the analysis. Different normalization methods have been used, such as square root (Smith and Emberlin, 2005), natural base logarithm (Alcázar et al., 2004) and base ten logarithm (Stach et al., 2008). Other models used include artificial neural network, autoregressive integrated model, logistic regression, partially linear and regression tree. As all these models do not require normality and linearity in the data, they can be implemented on the original data. Other models such as Box-Jenkins (using or not ARIMA model) were found but are not considered because it requires stable patterns that are not found in annual series of aerobiological data.

In all OBM it is possible to measure the degree of fit of the models. The most frequent fit measure used in the articles analyzed has been the coefficient R^2. Although this coefficient has been calculated for different sites, periods, and taxa, in the models applied to pollen, this coefficient showed a range from 0.21 to 0.99, while for the fungal spores it varied between 0.16 and 0.98. In both cases, models meeting data normality and linearity presented lower values of adjustment than those not requiring them (such as logistic regression, artificial neural network, between other). Additionally, and specifically for pollen, R^2 does not differ between models depending on the forecast parameter.

If the OBM are chronologically evaluated, it is noted that authors tend to decrease the use of the linear regression, while methods such as artificial neural network and logistic regression tend to be more frequently used. This can be attributed to two facts. Firstly, in recent decades these new tools have been applied to aerobiological data, and secondly, the fact that these new models are not affected by the absence of linearity and normality in aerobiological data. Furthermore, these computational methods can be used, amongst others, simultaneously for classification, numerical prediction or grouping (Scheifinger et al., 2013).

Table 4 shows the percentage distribution by type of method applied in OBM. As commented above, linear regression is the most applied model, followed by artificial neural network, logistic regression, and autoregressive integrated model. It is noteworthy that under the name of linear regression, other variations of the method are included such as multiple regression (32.7% of studies included in Linear regression), stepwise multiple regression (31.0%), backward multiple regression (4.6%) and multiple hierarchical regression (1.9%). Taking into account that in one paper more than one method and taxon may be used, in the 152 articles reviewed, within these OBM corresponded to 55.5%, being more frequently used in pollen studies (44.8%) and less frequently in papers related to fungal spores (10.7%).

The relationship between the model type used and the taxa analyzed (Table 5) shows that the most frequent pollen taxon is Poaceae (21.3% of the total models applied to pollen types), followed by *Olea* (20.8%), *Ambrosia* (9.8%) and *Betula* (7.5%) of a total of 28 pollen taxa. In the case of fungal spores, *Alternaria* is the most common (44.4% of the total models applied to fungal spores), followed by *Cladosporium* (22.2%) and *Ganoderma* (12.9%) of a total of seven taxa. The frequency of application of models for each taxon may be justified by the interest, importance, or impact of the taxon on health problems or on agronomic interests and not by the specific constraints with the methods.

Table 6 shows the relationship between the model and the parameter modelled. In the case of pollen, the most modelled variable is daily values, followed by the season start day, and the season end day and season duration with the same number of the cases. In the case of fungal spores, as in the pollen case, the daily values are the most modelled parameter.

2.2. PHenological Models (PHM)

PHM in this review include the aerobiological models based on the pollen season (e.g., season start, season end and season duration) that describe phenological phases. Due to the fact that airborne pollen and phenological events are traced back to the flowering of the plants, to understand when these events occur it is necessary to control the weather conditions. Many authors have forecasted the start date of the pollen season in different taxa using temperature (García-Mozo et al., 2000, 2002; Orlandi et al., 2002; Sánchez-Mesa et al., 2002; Recio et al., 2012). However, radiation or evapotranspiration insolation, sunshine hours, wind and rainfall were also applied (Durand, 1967; Mandrioli et al., 1993; Fornaciari et al., 1998; Laaidi, 2001a, 2001b; Laaidi et al., 2003; Stach et al., 2008; Kizilpinar et al., 2011; Puc, 2012). All these variables have been applied in the study of phenological phases (Chuine et al., 1998; Chuine, 2000; Schaber and Badeck, 2003; Linkosalo et al., 2008). As Scheifinger et al. (2013) showed, these models employed accumulation of heat (Sum temperature, Growing degree days, or Heat units), dormancy units (Chilling Unit), photoperiod, and water availability, even two or more of these parameters.

However, studies relating airborne pollen concentrations with PHM are not abundant, although Khwarahm et al. (2014) asserted that the meteorological variables affect the start of flowering and the pre-peak period, as well as the end flowering and post-peak period. Andersen (1991) is one of the first papers in which PHM were used in aerobiological studies, using "Chilling units" and "Growing degree hours" to predict the start of the pollen season of *Alnus*, *Ulmus* and *Betula* in Denmark.

In this review, PHM are represented by 24 different models (Table 2). One model used the accumulation heat and photoperiod ("Forc CO"); three models used the dormancy phase as main parameter ("Chilling day", "Chilling hour" and "Chilling unit"); eight used the accumulation heat and dormancy phase ("Alternative", "Parallel", "Sequential", "Parallel Sarvas", "Parallel thermal time", "Sequential Sarvas",

Table 1
Observation Based Models (OBM) applied to the study of airborne pollen and spore since 1998–2016.

Model	Taxon	Forecasting parameter	Reference
Artificial neural network	*Alternaria*	Daily values	(Astray et al., 2010) (Bruno et al., 2007) (Grinn-Gofroń and Strzelczak, 2008a) (Grinn-Gofroń and Strzelczak, 2011) (Grinn-Gofroń and Strzelczak, 2013) (Tomassetti et al., 2009) (Tomassetti et al., 2013)
		Hourly values	(Grinn-Gofroń and Strzelczak, 2009)
		Spatial estimation	(Tomassetti et al., 2013)
	Ambrosia	Daily values	(Csépe et al., 2014)
	Betula	Daily values	(Puc, 2012)
	Castanea	Daily values	(Astray et al., 2016)
	Cladosporium	Daily values	(Grinn-Gofroń and Strzelczak, 2008a) (Grinn-Gofroń and Strzelczak, 2011) (Grinn-Gofroń and Strzelczak, 2013)
		Hourly values	(Grinn-Gofroń and Strzelczak, 2009)
	Fungal spores[e]	Values	(Verma and Pathak, 2009)[a]
	Ganoderma	Daily values	(Jedryczka et al., 2015) (Sadyś et al., 2016b)
		Hourly values	(Kasprzyk et al., 2011)
		Present or absence	(Grinn-Gofroń and Strzelczak, 2011)
	Olea	APIn	(Oteros et al., 2013b)
		Daily values	(Aznarte et al., 2007) (Iglesias-Otero et al., 2015a) (Voukantsis et al., 2010a) (Voukantsis et al., 2010b)
		No. days reach threshold	(Oteros et al., 2013b)
		Peak values	(Oteros et al., 2013b)
		Season start day	(Oteros et al., 2013b)
	Plantago	Daily values	(Iglesias-Otero et al., 2015b)
	Pleospora	Daily values	(Bruno et al., 2007) (Tomassetti et al., 2009) (Tomassetti et al., 2013)
		Spatial estimation	(Tomassetti et al., 2013)
	Poaceae	Daily values	(Arca et al., 2004) (Rodríguez-Rajo et al., 2010a) (Sánchez-Mesa et al., 2002) (Sánchez-Mesa et al., 2005) (Voukantsis et al., 2010a) (Voukantsis et al., 2010b)
	Pollen[e]	Daily values	(Ranzi et al., 2003)
	Pyricularia	Numbers of spores	(Mojerlou et al., 2013)
	Urticaceae	Daily values	(Dara, 2013)[b] (Voukantsis et al., 2010b)
Autoregressive integrated	*Alnus*	Daily values	(Rodríguez-Rajo et al., 2006)
	Alternaria	Daily values	(Damialis and Gioulekas, 2006) (Escuredo et al., 2011)
		MPIn	(Damialis and Gioulekas, 2006)
	Ambrosia	Daily values	(Matyasovszky and Makra, 2011)
	Betula	Daily values	(Castellano-Méndez et al., 2005)[c]
	Botrytis	Daily values	(Fernández-González et al., 2012) (Rodríguez-Rajo et al., 2010b)
	Castanea	Daily values	(Rodríguez-Rajo et al., 2005b)
	Cladosporium	Daily values	(Damialis and Gioulekas, 2006)
		MPIn	
	Cupressaceae	Daily values	(Ocaña-Peinado et al., 2013)
	Fungal spores[e]	Values	(Verma and Pathak, 2009)[a]
	Olea	APIn	(García-Mozo et al., 2014)
		Season duration	
		Season start day	
	Poaceae	Daily values	(Tassan-Mazzocco et al., 2015)
		Season start day	
		Weekly values	
	Urticaceae	Daily values	(Tassan-Mazzocco et al., 2015)
		Season start day	
		Weekly values	
Gamma PDF	*Alternaria*	Season end day	(Kasprzyk and Walanus, 2014)
		Season duration	
		Season start day	
	Artemisia	Season end day	(Kasprzyk and Walanus, 2014)
		Season duration	
		Season start day	
Gaussian PDF	*Alternaria*	Season end day	(Kasprzyk and Walanus, 2014)
		Season duration	
		Season start day	
	Artemisia	Season end day	(Kasprzyk and Walanus, 2014)
		Season duration	
		Season start day	
Gumbel's PDF	*Ambrosia*	No. days reach threshold	(Puc and Wolski, 2013)
		Peak day	
		Peak values	
		Season duration	
		Season start day	
	Artemisia	No. days reach threshold	(Puc and Wolski, 2013)
		Peak day	
		Peak values	
		Season duration	
		Season start day	
	Poaceae	No. days reach threshold	(Puc and Wolski, 2013)
		Peak day	
		Peak values	
		Season duration	
		Season start day	
Linear discriminant	*Olea*	APIn	(Oteros et al., 2013b)
		No. days reach threshold	
		Peak values	
		Season start day	

Table 1 (*continued*)

Model	Taxon	Forecasting parameter	Reference
	Poaceae	Daily values	(Sánchez-Mesa et al., 2005)
Linear regression	*Alnus*	Peak day	(Piotrowska-Weryszko, 2013)
		Season duration	
		Season start day	(Emberlin et al., 2007) (Piotrowska-Weryszko, 2013) (Rodríguez-Rajo et al., 2009)
	Alternaria	Daily values	(Aira et al., 2008) (Recio et al., 2012) (Rodríguez-Rajo et al., 2005a) (Sidel et al., 2015)
		Weekly values	(Recio et al., 2012)
	Ambrosia	Daily values	(Howard and Levetin, 2014)[b] (Makra and Matyasovszky, 2011) (Makra et al., 2004)
		Season duration	(Laaidi et al., 2003) (Zhang et al., 2015)
		Season start day	(Laaidi et al., 2003) (Zhang et al., 2015)
	Artemisia	APIn	(Drzeniecka-Osiadacz et al., 2015)[a]
		Season duration	(Zhang et al., 2015)
		Season start day	(Drzeniecka-Osiadacz et al., 2015) (Zhang et al., 2015)
	Betula	Daily values	(Méndez et al., 2005)[b]
		Season end day	(Myszkowska, 2013)
		Peak day	
		Peak values	
		Season duration	(Zhang et al., 2015)
		Season start day	(Adams-Groom et al., 2002) (Gormsen et al., 2005) (Laaidi, 2001b) (Myszkowska, 2013) (Zhang et al., 2015)
	Cheno.-Amara.	Daily values	(Angosto et al., 2005)[b]
	Cladosporium	Daily values	(Aira et al., 2008) (Angulo-Romero et al., 1999) (Molina et al., 1998) (Recio et al., 2012) (Rodríguez-Rajo et al., 2006)
		Weekly values	(Recio et al., 2012)
	Corylus	Season start day	(Emberlin et al., 2007) (Piotrowska and Kaszewski, 2009)
	Cupressaceae	APIn	(Galán et al., 1998a)
		Daily values	(Díaz de la Guardia et al., 2006) (Galán et al., 1998b) (Ocaña-Peinado et al., 2013) (Sabariego et al., 2012)[b]
		Peak day	(Galán et al., 1998a)
		Peak values	
		Season start day	
	Fungal spores[e]	Values	(Ho et al., 2005)[a,d] (Verma and Pathak, 2009)[a]
	Olea	APIn	(Galán et al., 2001a) (Oteros et al., 2013a) (Rojo et al., 2016) (Sicard et al., 2012)
		Daily values	(Díaz de la Guardia et al., 2003) (Voukantsis et al., 2010a) (Voukantsis et al., 2010b)
		Season end day	(Murray and Galán, 2016) (Sicard et al., 2012)
		Peak day	(Galán et al., 2001a) (Murray and Galán, 2016)
		Peak values	(Sicard et al., 2012)
		Pre-peak pollen integral	(Sicard et al., 2012)
		Season duration	(Murray and Galán, 2016) (Sicard et al., 2012)
		Season start day	(Fornaciari et al., 1998) (Murray and Galán, 2016) (Ribeiro et al., 2006) (Sicard et al., 2012)
	Platanus	Daily values	(Iglesias et al., 2007) (Sabariego et al., 2008)
		Season start day	(Iglesias et al., 2007)
	Poaceae	APIn	(Emberlin et al., 1999) (Schäppi et al., 1998)
		Daily values	(Aboulaich et al., 2013) (Matyasovszky et al., 2011) (Smith and Emberlin, 2005) (Smith and Emberlin, 2006) (Stach et al., 2008) (Toro et al., 1998)[b] (Voukantsis et al., 2010a) (Voukantsis et al., 2010b)
		Season end day	(Kasprzyk and Walanus, 2010) (Stach et al., 2008)
		Peak day	(Kasprzyk and Walanus, 2010) (Piotrowska, 2012) (Stach et al., 2008)
		Peak values	(Piotrowska, 2012)
		Season duration	(Kasprzyk and Walanus, 2010) (Piotrowska, 2012) (Zhang et al., 2015)
		Season start day	(Kasprzyk and Walanus, 2010) (Laaidi, 2001a) (Piotrowska, 2012) (Stach et al., 2008) (Zhang et al., 2015)
	Pollen[e]	Daily values	(Angosto et al., 2005)[b]
	Prunus	Season start day	(Gormsen et al., 2005)
	Quercus	Season duration	(Zhang et al., 2015)
		Season start day	
	Urticaceae	Daily values	(Dara, 2013)[b] (Galán et al., 2000) (Voukantsis et al., 2010b)
Logistic regression	*Acer*	Season end day	(Ribeiro et al., 2007b)
		Season start day	
	Alnus	Daily values	(Myszkowska and Majewska, 2014)[c]
		Season end day	(Ribeiro et al., 2007b)
		Season start day	(Myszkowska, 2014a) (Ribeiro et al., 2007b)
	Alternaria	Annual increase	(De Linares et al., 2010)
	Betula	Daily values	(Cotos-Yáñez et al., 2004) (Myszkowska and Majewska, 2014)[c]
		Season start day	(Myszkowska, 2014a)
	Castanea	Season end day	(Ribeiro et al., 2007b)

Table 1 (*continued*)

Model	Taxon	Forecasting parameter	Reference
		Season start day	
	Corylus	Daily values	(Myszkowska and Majewska, 2014)[c]
		Season start day	(Myszkowska, 2014a)
	Cupressaceae	Season end day	(Ribeiro et al., 2007b)
		Season start day	
	Fraxinus	Season end day	(Ribeiro et al., 2007b)
		Season start day	
	Olea	Weekly values	(Escabias et al., 2013)
	Pinus	Season end day	(Ribeiro et al., 2007b)
		Season start day	
	Platanus	Season end day	(Ribeiro et al., 2007b)
		Season start day	
	Poaceae	Daily values	(Myszkowska and Majewska, 2014)[c]
		Season end day	(Ribeiro et al., 2007b)
		Season start day	(Myszkowska, 2014a) (Ribeiro et al., 2007b)
	Urticaceae	Season end day	(Ribeiro et al., 2007b)
		Season start day	
Nonparametric median regression	*Ambrosia*	Daily values	(Makra and Matyasovszky, 2011)
Nonparametric regression	*Ambrosia*	Daily values	(Makra and Matyasovszky, 2011) (Makra et al., 2011) (Makra et al., 2011)[c]
		Season end day	(Makra et al., 2011)
		Season start day	
Partial least squares	*Olea*	Peak day	(Aguilera et al., 2015)
		Season start day	
	Poaceae	Daily values	(Brighetti et al., 2014)
Partially linear Poisson regression	*Betula*	Daily values	(Cotos-Yáñez et al., 2004)
	Poaceae	Daily values	(Erbas et al., 2007)
Principal component analysis	Cupressaceae	Daily values	(Ocaña-Peinado et al., 2008)
Regression tree	*Ambrosia*	Daily values	(Csépe et al., 2014)
	Ganoderma	Daily values	(Jedryczka et al., 2015) (Sadyś et al., 2016a)
		Present or absence	(Grinn-Gofroń and Strzelczak, 2011)
	Olea	APIn	(Oteros et al., 2013b)
		Daily values	(Voukantsis et al., 2010b)
		No. days reach threshold	(Oteros et al., 2013b)
		Peak values	
		Season start day	
	Platanus	Daily values	(Sabariego et al., 2008)
	Poaceae	Daily values	(Voukantsis et al., 2010b)
	Urticaceae	Daily values	(Voukantsis et al., 2010b)
Shuffle complex evolution metropolis algorithm	Cupressaceae	Daily values	(Silva-Palacios et al., 2016)
	Olea	Daily values	(Fernández-Rodríguez et al., 2016a)
	Quercus	Daily values	(Fernández-Rodríguez et al., 2016b)
Stochastic gradient boosting	*Alnus* *Ambrosia* *Artemisia* *Betula* *Carpinus* *Castanea* *Corylus* *Fagus* *Fraxinus* *Plantago* *Platanus* Poaceae *Populus* *Quercus* *Rumex* Urticaceae	Daily values	(Hilaire et al., 2012)
Support vector machine	*Olea* Poaceae Urticaceae	Daily values	(Voukantsis et al., 2010b)

PDF: probability distribution function. Cheno.-Amara.: Chenopodiaceae-Amaranthaceae. APIn: Annual pollen integral. MPIn: Month pollen integral.
[a] Not reported time of integration.
[b] Standardized concentration.
[c] Concentration by threshold.
[d] 15 different taxa.
[e] No reported the taxon of pollen or spore.

"Sequential thermal time" and "Forcing and chilling units"); and the other 12 models used the accumulation heat. Table 4 shows that amongst the predictive models analyzed, PHM corresponds to 38.5% of the total, with "Growth degree days (°GDD)" being the most frequent one, followed by "Parallel" and "Sequential" group models and, "Alternative" and "Spring warming" models.

The complexity of PHM is to establish the characteristics of the season (the season start and end) by means of temperature thresholds for each phenological phase. In fact, the method to define these thresholds is more complex than the application of the model itself. Analyzing the articles selected in this review, 24 out of 33 studies varied the threshold between 0 and 12 °C in order to optimize the model and, 6 of these 33 studies used a methodology that crosses the variation of temperature threshold with the way the season start was determined.

As in OBM, in PHM it is possible to measure the degree of fit between the models and reality. The fit applied with most frequency in the articles analyzed was RMSE (Root Mean Square Error). The results showed a good fit of PHM with a range of 0 to 7 days for season start predictions, and of 1 to 6 days for the prediction of the peak day. The validations in all the studies showed that these PHM predict the modelled data a few days after than original data.

The analysis of the taxa studied with PHM (Table 5) shows that *Olea* is the taxon more frequently studied, followed by *Alnus*, *Platanus* and *Quercus*. Like in the case of OBM, the study of a genus or other is not due to a specific condition of the model, but to the importance or impact of the taxon on health problems or on agronomic interest. Finally, Table 6 shows that the season start day is the most modelled, followed by season duration and peak day.

2.3. Other models (OTM)

Under this name, we present the aerobiological predictive models that do not fit with the previous models defined. Table 3 shows nine types of models. The "bioclimatic index" and "taxonomic index" are models that predict the characteristics of the season. These models simplify the input variables (meteorological and phenological conditions) through indexes. Valencia-Barrera et al. (2002) showed that in the short-term (taking account only main pollen/spore season) the use of the "Bioclimatic index" model shows similar results than OBM.

Table 2
Phenological Based Models (PHM) applied to the study of airborne pollen and spore since 1998–2016.

Model	Taxon	Forecasting parameter	Reference
Alternative	*Acer*	Season start day	(Siniscalco et al., 2015)
	Aesculus	Season start day	(Chuine et al., 1998)
	Alnus		
	Alnus	Season start day	(Siniscalco et al., 2015)
	Ambrosia		
	Artemisia		
	Buxus	Season start day	(Chuine et al., 1998)
	Carpinus	Season start day	(Siniscalco et al., 2015)
	Castanea		
	Juglans		
	Olea	Season start day	(Chuine et al., 1998)
	Platanus		
	Platanus	Season start day	(Siniscalco et al., 2015)
	Taxus	Season start day	(Chuine et al., 1998)
	Ulmus		
Chilling day	*Alnus*	Season start day	(Novara et al., 2016)
	Corylus		
Chilling hours	*Alnus*	Season start day	(González-Parrado et al., 2006)
	Cupressaceae	Season start day	(Fuertes-Rodríguez et al., 2007)
	Olea	Season start day	(Rodríguez-Rajo et al., 2004a)
	Quercus	Season start day	(García-Mozo et al., 2000) (Rodríguez-Rajo et al., 2003) (Rodríguez-Rajo et al., 2005c)
Chilling unit	*Olea*	Peak day	(Orlandi et al., 2006)
Double-threshold temperature sum	*Betula*	Season duration	(Linkosalo et al., 2010)
Forc PO	*Olea*	Peak day	(García-Mozo et al., 2009)
		Season start day	
Forcing and chilling units	*Alnus*	Season start day	(Pauling et al., 2014)
	Betula		
	Corylus		
	Fraxinus		
	Poaceae		
	Quercus	Peak day	(García-Mozo et al., 2008)
		Season start day	
Forcing Sarvas	*Aesculus*	Season start day	(Chuine et al., 1999)
	Alnus		
	Betula		
	Buxus		
	Carpinus		
	Castanea		
	Corylus		
	Olea	Season start day	(Chuine et al., 1999) (Osborne et al., 2000)
	Platanus	Season start day	(Chuine et al., 1999)
	Taxus		
	Ulmus		
Forcing thermal time	*Aesculus*	Season start day	(Chuine et al., 1999)
	Alnus		
	Betula		
	Buxus		
	Carpinus		
	Castanea		
	Corylus		
	Olea	Season start day	(Chuine et al., 1999) (Osborne et al., 2000)
	Platanus	Season start day	(Chuine et al., 1999)
	Taxus		
	Ulmus		
Grow degree days (°GDD)	*Alnus*	Season start day	(Frenguelli and Bricchi, 1998) (González-Parrado et al., 2006)
	Ambrosia	Season duration	(Zhang et al., 2015)
		Season start day	
	Artemisia	Season duration	
		Season start day	
	Betula	Season duration	
		Season start day	
	Corylus	Season start day	(Frenguelli and Bricchi, 1998)
	Cupressaceae	Season end day	(Malaspina et al., 2007)
		Season start day	(Fuertes-Rodríguez et al., 2007)
		Season start day	(Malaspina et al., 2007)
	Olea	Peak day	(Achmakh et al., 2015)
		Season start day	(Achmakh et al., 2015) (Galán et al., 2001b) (Galán et al., 2005)
	Platanus	Season start day	(Alcázar et al., 2011)
	Poaceae	Season duration	(Zhang et al., 2015)
		Season start day	
	Populus	Season start day	(Frenguelli and Bricchi, 1998)
	Quercus	Season duration	(Zhang et al., 2015)
		Season start day	(García-Mozo et al., 2000) (García-Mozo et al., 2006) (Kasprzyk, 2009) (Rodríguez-Rajo et al., 2004b) (Zhang et al., 2015)
	Salix	Season start day	(Frenguelli and Bricchi, 1998)
	Ulmus		
Grow degree hours (°GDH)	*Betula*	Season duration	(Zhang et al., 2014)
		Season start day	
	Quercus	Season duration	(Zhang et al., 2014)
		Season start day	
Heat units	*Olea*	Season start day	(Galán et al., 2001b)
	Quercus	Season start day	(García-Mozo et al., 2002)
Parallel	*Acer*	Season start day	(Siniscalco et al., 2015)
	Aesculus	Season start day	(Chuine et al., 1998)
	Alnus	Season start day	(Chuine et al., 1998) (Siniscalco et al., 2015)
	Ambrosia	Season start day	(Siniscalco et al., 2015)
	Artemisia		
	Buxus	Season start day	(Chuine et al., 1998)
	Carpinus	Season start day	(Siniscalco et al., 2015)
	Castanea		
	Juglans		
	Olea	Season start day	(Chuine et al., 1998)
	Platanus	Season start day	(Chuine et al., 1998) (Siniscalco et al., 2015)
	Taxus	Season start day	(Chuine et al., 1998)
	Ulmus		
Parallel Sarvas	*Aesculus*	Season start day	(Chuine et al., 1999)
	Alnus		
	Betula		
	Buxus		
	Carpinus		
	Castanea		
	Corylus		

Table 2 (*continued*)

Model	Taxon	Forecasting parameter	Reference
	Olea		
	Platanus		
	Taxus		
	Ulmus		
Parallel thermal time	*Aesculus*	Season start day	(Chuine et al., 1999)
	Alnus		
	Betula		
	Buxus		
	Carpinus		
	Castanea		
	Corylus		
	Olea		
	Platanus		
	Taxus		
	Ulmus		
Q10	Poaceae	Season start day	(Laaidi, 2001a)
Sequential	*Acer*	Season start day	(Siniscalco et al., 2015)
	Aesculus	Season start day	(Chuine et al., 1998)
	Alnus	Season start day	(Chuine et al., 1998) (Siniscalco et al., 2015)
	Ambrosia	Season start day	(Siniscalco et al., 2015)
	Artemisia		
	Buxus	Season start day	(Chuine et al., 1998)
	Carpinus	Season start day	(Siniscalco et al., 2015)
	Castanea		
	Juglans		
	Olea	Season start day	(Chuine et al., 1998)
	Platanus	Season start day	(Chuine et al., 1998) (Siniscalco et al., 2015)
	Taxus	Season start day	(Chuine et al., 1998)
	Ulmus		
Sequential Sarvas	*Aesculus*	Season start day	(Chuine et al., 1999)
	Alnus		
	Betula		
	Buxus		
	Carpinus		
	Castanea		
	Corylus		
	Olea		
	Platanus		
	Taxus		
	Ulmus		
Sequential thermal time	*Aesculus*	Season start day	(Chuine et al., 1999)
	Alnus		
	Betula		
	Buxus		
	Carpinus		
	Castanea		
	Corylus		
	Olea		
	Platanus		
	Taxus		
	Ulmus		
Spring warming	*Acer*	Season start day	(Siniscalco et al., 2015)
	Aesculus	Season start day	(Chuine et al., 1998)
	Alnus	Season start day	(Chuine et al., 1998) (Siniscalco et al., 2015)
	Ambrosia	Season start day	(Siniscalco et al., 2015)
	Artemisia		
	Buxus	Season start day	(Chuine et al., 1998)
	Carpinus	Season start day	(Siniscalco et al., 2015)
	Castanea		
	Juglans		
	Olea	Season start day	(Chuine et al., 1998)
	Platanus	Season start day	(Chuine et al., 1998) (Siniscalco et al., 2015)
	Taxus	Season start day	(Chuine et al., 1998)
	Ulmus	Season start day	(Chuine et al., 1998)
State of forcing function	Poaceae	Peak day	(García-Mozo et al., 2009)
		Season start day	
Temperature sum	*Ambrosia*	Season duration	(Laaidi et al., 2003)
		Season start day	
	Betula	Season start day	(Laaidi, 2001b)
	Olea	Season start day	(Galán et al., 2001b)
	Plantago	Season start day	(González-Parrado et al., 2014)
	Poaceae	Season start day	(Laaidi, 2001a) (Myszkowska, 2014b)
	Quercus	Season start day	(García-Mozo et al., 2000)[a]
Thermal time	*Olea*	Season start day	(Osborne et al., 2000)
Unified forcing	*Olea*	Season start day	(Osborne et al., 2000)

[a] The method has two variations, with the average of the maximum temperatures and the sum of the maximum temperature.

However, in the long term (taking the period of pre and post main pollen/spore season), the Bioclimatic index shows a higher correlation, allowing to conclude that these indices could offer a more precise prediction. Alternatively, the "Daily mean-based forecast" model provides a daily estimate based on the mean of historical pollen concentration of the same date (day of the year), as a "pollen calendar".

The rest of models employ statistical interpolation methods with data from nearby stations to the area of interest. DellaValle et al. (2012) applied three models in the same study: kriging, longitudinal and near neighbor. They concluded that these estimations may be unreliable due to the influence of meteorological conditions, the aerodynamic characteristics of the particle and the high variations in measurements but that they could provide good estimations in unmonitored areas. On the other hand, Alba et al. (2006) applied kriging to estimate *Olea* pollen concentrations in the Andalucía region (Spain). In this case, they did not find significant differences between observed values and model estimates on the average weekly pollen concentrations, except when monitoring stations showed a high fluctuation in the concentrations.

These model types correspond to 6.0% of the papers analyzed (Table 4). The most popular model was "kriging", followed by "daily mean-based forecast", "longitudinal models" and "the nearest monitor method". Table 5 shows that the most frequently modelled taxon was Poaceae, followed by *Quercus*, *Acer* and Urticaceae. In the case of spores, there was only one study for *Alternaria* and *Cladosporium* taxa. Finally, Table 6 shows that the most frequently modelled parameter was "Spatial estimation" followed by "Season start".

3. Conclusion

Airborne pollen and fungal spore modeling have a progressively increased over recent years, evidencing the utility of these investigations in diverse transverse fields with aerobiological applications. In general, it was observed that the predictive models were applied to a wide number of aerobiological parameters. However, there is a clear predominance of using OBM to predict future concentrations, mostly daily values, while PHM were used to characterize the season, mostly at the start of the pollen season. Likewise, OTM focuses on establishing spatial

Table 3
Other models (OTM) applied in the study of airborne pollen and spore since 1998–2016.

Model	Taxon	Forecasting parameter	Reference
Bioclimatic index	*Quercus*	Season start day	(Kasprzyk, 2009)
Daily mean-based forecast	Poaceae	Daily values Season start day Weekly values	(Tassan-Mazzocco et al., 2015)
	Urticaceae	Daily values Season start day Weekly values	
Indicator taxa	Pollen	Season duration	(Valencia-Barrera et al., 2002)
	Quercus	Season start day	(Kasprzyk, 2009)
Kriging	*Acer*	Spatial estimation	(DellaValle et al., 2012)
	Olea	Spatial estimation	(Alba et al., 2006)
	Poaceae *Quercus* Tree pollen Weed pollen	Spatial estimation	(DellaValle et al., 2012)
Kriging-linear regression-spatial autocorrelation	*Olea*	Spatial estimation	(Rojo et al., 2016)
Land used regression	*Asper.-Penic* *Cladosporium* Fungal spores	Spatial estimation	(Kallawicha et al., 2015)
Longitudinal	*Acer* Poaceae *Quercus* Tree pollen Weed pollen	Spatial estimation	(DellaValle et al., 2012)
Near neighbor	*Acer*	Spatial estimation	(DellaValle et al., 2012)
	Ambrosia	Daily values	(Matyasovszky and Makra, 2012)[a]
	Poaceae *Quercus* Tree pollen Weed pollen	Spatial estimation	(DellaValle et al., 2012)

[a] Concentration estimated by thresholds. Asper.-Penic.: *Aspergillus-Penicillium.*

Table 4
Distribution of forecasting models of airborne pollen and spores. ($n = 503$, studies since 1998–2016). Italic is to highlight subtotal by model type.

Model		Percentage of used		
		Pollen	Fungal spores	Total
OBM	Artificial neural network	4.2%	4.6%	8.7%
	Autoregressive integrated	2.8%	1.6%	4.4%
	Gamma PDF	0.6%	0.6%	1.2%
	Gaussian PDF	0.6%	0.6%	1.2%
	Gumbel's PDF	3.0%		3.0%
	Linear discriminant	1.0%		1.0%
	Linear regression	18.5%	2.6%	21.1%
	Logistic regression	5.6%	0.2%	5.8%
	Nonparametric median regression	0.2%	0.0%	0.2%
	Nonparametric regression	1.0%		1.0%
	Partial least squares	0.6%		0.6%
	Partially linear	0.2%		0.2%
	Poisson regression	0.2%		0.2%
	Principal component analysis	0.2%		0.2%
	Regression tree	1.8%	0.6%	2.4%
	Shuffle Complex Evolution Metropolis Algorithm (SCEM-UA)	0.6%		0.6%
	Stochastic gradient boosting	3.2%		3.2%
	Support vector machine	0.6%		0.6%
	Subtotal of OMB	*44.8%*	*10.7%*	*55.5%*
PHM	Alternative	3.0%		3.0%
	Chilling days	0.4%		0.4%
	Chilling hours	1.2%		1.2%
	Chilling units	0.2%		0.2%
	Double-threshold temperature sum	0.2%		0.2%
	Forc PO	0.4%		0.4%
	Forcing and chilling units	1.4%		1.4%
	Forcing Sarvas	2.4%		2.4%
	Forcing thermal time	2.4%		2.4%
	Growing degree days	5.6%		5.6%
	Growing degree hours	0.8%		0.8%
	Heat units	0.4%		0.4%
	Parallel	3.0%		3.0%
	Parallel Sarvas	2.2%		2.2%
	Parallel thermal time	2.2%		2.2%
	Q10	0.2%		0.2%
	Sequential	3.0%		3.0%
	Sequential Sarvas	2.2%		2.2%
	Sequential thermal time	2.2%		2.2%
	Spring warming	3.0%		3.0%
	State of forcing function	0.4%		0.4%
	Temperature sum	1.6%		1.6%
	Thermal time	0.2%		0.2%
	Unified forcing	0.2%		0.2%
	Subtotal of PHM	*38.5%*		*38.5%*
OTM	Bioclimatic index	0.4%		0.4%
	Daily mean-based forecast	1.2%		1.2%
	Kriging	1.2%		1.2%
	Kriging-linear regression-spatial autocorrelation (Kr + Lr + Sa)	0.2%		0.2%
	Land used regression		0.6%	0.6%
	Longitudinal	1.0%		1.0%
	Near neighbor	0.2%		0.2%
	Taxonomic indicator	0.2%		0.2%
	The nearest monitor method	1.0%		1.0%
	Subtotal of others	*5.4%*	*0.6%*	*6.0%*
Total		*88.7%*	*11.3%*	

OBM: Observation Based Models. PHM: Phenological Based Models. OTM: other models.

estimations in unmonitored areas. In the particular case of OBM, it was observed that the frequency of statistical methods require the fulfillment of normality and linearity have been decreasing while those that do not require it have been increasing.

A greater amount of diversity was reported for PHM. However, many of these models had the same mathematical basis, and only the equation or criterion by which the development of the equation was established varied. Additionally, PHM used a smaller number of independent variables, generally the temperature, while OBM used the temperature and the precipitation.

Finally, it was possible to show that although PHM are simpler to apply than OBM, the complexity in the definition of the start point criteria in the PHM generates wide differences in the results obtained, so researchers need to include more detailed discussions to arrange these aspects in the models.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 5
Forecasting models applied by taxon (more than one method may be used in one paper). Italic numbers are to highlight subtotal by model type or by type of bioaerosol.

Model		Pollen taxa																			
		Acer	*Aesculus*	*Alnus*	*Ambrosia*	*Artemisia*	*Betula*	*Buxus*	*Carpinus*	*Castanea*	Chen.-Amar.	*Corylus*	Cupressaceae	*Fagus*	*Fraxinus*	*Juglans*	*Olea*	*Pinus*	*Plantago*	*Platanus*	Poaceae
OBM	Artificial neural network				1		1			1							8		1		6
	Autoregressive integrated			1	1		1			1			1				3				3
	Gamma PDF					3															
	Gaussian PDF					3															
	Gumbel's PDF				5	5															5
	Linear discriminant																4				1
	Linear regression			5	7	4	10				1	2	8				22			3	24
	Logistic regression	2		4			3			2		2	2		2		1	2		2	4
	Nonparametric median regression				1																
	Nonparametric regression				5																
	Partial least squares																2				1
	Partially linear						1														
	Poisson regression																				1
	Principal component analysis												1								
	Regression tree				1												5			1	1
	SCEM-UA												1				1				
	Stochastic gradient boosting			1	1	1	1		1	1		1		1	1				1	1	1
	Support vector machine																1				1
	Subtotal of OBM	*2*		*11*	*22*	*16*	*17*		*1*	*5*	*1*	*5*	*13*	*1*	*3*		*47*	*2*	*2*	*7*	*48*
PHM	Alternative	1	1	2	1	1		1	1	1						1	1			2	
	Chilling day			1								1									
	Chilling hours			1									1				1				
	Chilling unit																1				
	Double-threshold temperature sum						1														
	Forc PO																2				
	Forcing and chilling units			1			1					1			1						1
	Forcing Sarvas		1	1			1	1	1	1		1					2			1	
	Forcing thermal time		1	1			1	1	1	1		1					2			1	
	°GDD			2	2	2	2					1	3				4			1	2
	°GDH						2														
	Heat units																1				
	Parallel	1	1	2	1	1		1	1	1						1	1			2	
	Parallel Sarvas		1	1			1	1	1	1		1					1			1	
	Parallel thermal time		1	1			1	1	1	1		1					1			1	
	Q10																				1
	Sequential	1	1	2	1	1		1	1	1						1	1			2	
	Sequential Sarvas		1	1			1	1	1	1		1					1			1	
	Sequential thermal time		1	1			1	1	1	1		1					1			1	
	Spring warming	1	1	2	1	1		1	1	1						1	1			2	
	State of forcing function																				2
	Temperature sums				2		1										1		1		2
	Thermal time																1				
	Unified forcing																1				
	Subtotal of PHM	*4*	*10*	*19*	*8*	*6*	*13*	*10*	*10*	*10*		*9*	*4*		*1*	*4*	*24*		*1*	*15*	*8*
OTM	Bioclimatic index																				
	Daily mean-based forecast																				3
	Kriging	1															1				1
	Kriging + Lr + Sa																1				
	Land used regression																				
	Longitudinal	1																			1
	Near neighbor				1																
	Taxonomic index																				
	The nearest monitor method	1																			1
	Subtotal of other	*3*			*1*												*2*				*6*
Total		*9*	*10*	*30*	*31*	*22*	*30*	*10*	*11*	*15*	*1*	*14*	*17*	*1*	*4*	*4*	*73*	*2*	*3*	*22*	*62*

Table 5 *(continued)*

Model	Pollen taxa										Fungal spore taxa									Total by model
	Populus	*Prunus*	*Quercus*	*Rumex*	*Salix*	*Taxus*	*Ulmus*	Urticaceae	Pollen	Subtotal	*Alternaria*	*Asper.-Pen.*	*Botrytis*	*Cladosporium*	*Ganoderma*	*Pleospora*	*Pyricularia*	Fungal spores	Subtotal	
OBM								2	1	*21*	9			4	4	4	1	1	*23*	*44*
								3		*14*	3		2	2				1	*8*	*22*
										3	3								*3*	*6*
										3	3								*3*	*6*
										15										*15*
										5										*5*
		1	2					3	1	*93*	5			6				2	*13*	*106*
								2		*28*	1								*1*	*29*
										1										*1*
										5										*5*
										3										*3*
										1										*1*
										1										*1*
										1										*1*
								1		*9*					3				*3*	*12*
			1							*3*										*3*
	1		1	1				1		*16*										*16*
								1		*3*										*3*
	1	*1*	*4*	*1*				*13*	*2*	*225*	*24*		*2*	*12*	*7*	*4*	*1*	*4*	*54*	*279*
PHM						1	1			*15*										15
										2										2
			3							*6*										6
										1										1
										1										1
										2										2
			2							*7*										7
						1	1			*12*										12
						1	1			*12*										12
	1		6		1		1			*28*										28
			2							*4*										4
			1							*2*										2
						1	1			*15*										15
						1	1			*11*										11
						1	1			*11*										11
										1										1
						1	1			*15*										15
						1	1			*11*										11
						1	1			*11*										11
						1	1			*15*										15
										2										2
			1							*8*										8
										1										1
										1										1
	1		*15*		*1*	*10*	*11*			*194*										*194*
OTM			1						1	*2*										2
								3		*6*										6
			1						2	*6*										6
										1										1
												1		1				1	*3*	3
			1						2	*5*										5
										1										1
			1							*1*										1
			1						2	*5*										5
			5					*3*	*7*	*27*		*1*		*1*				*1*	*3*	*30*
Total	*2*	*1*	*24*	*1*	*1*	*10*	*11*	*16*	*9*	*446*	*24*	*1*	*2*	*13*	*7*	*4*	*1*	*5*	*57*	*503*

Cheno.-Amara.: Chenopodiaceae-Amaranthaceae. Asper.-Penic.: *Aspergillus-Penicillium*. SCEM-UA: Shuffle Complex Evolution Metropolis Algorithm. °GDD: Grow degree days. °GDH: Grow degree hours. Kriging + Lr + Sa: kriging-linear regression-spatial autocorrelation. OBM: Observation Based Models. PHM: Phenological Based Models. OTM: other models.

Table 6
Forecasting models applied by modelled parameter (more than one method may be used in one paper). Italic is to highlight subtotal by model type

Models		Parameter applied to pollen taxa											Parameter applied to spore taxa											
		APIn	Daily values	No. day reach threshold	Peak day	Peak value	Pre-Peak pollen integral	Season duration	Season end day	Season start day	Spatial estimation	Weekly values	Annual increase	Daily values	Hourly values	MPIn	Number of spores	Present or Absence	Season duration	Season end day	Season start day	Spatial estimation	Values[a]	Weekly values
OBM	Artificial neural network	1	16	1		1				1				15	3		1	1				2	1	
	Autoregressive integrated	1	8					1		3		2		5		2							1	
	Gamma PDF							1	1	1									1	1	1			
	Gaussian PDF							1	1	1									1	1	1			
	Gumbel's PDF			3	3	3		3		3														
	Linear discriminant	1	1	1		1				1														
	Linear regression	8	26		9	6	1	11	5	27				9									2	2
	Logistic regression		5						9	13		1	1											
	Nonparametric median regression		1																					
	Nonparametric regression		3						1	1														
	Partial least squares		1		1					1														
	Partially linear		1																					
	Poisson regression		1																					
	Principal component analysis		1																					
	Regression tree	1	5	1		1				1				2				1						
	SCEM-UA		3																					
	Stochastic gradient boosting		16																					
	Support vector machine		3																					
	Subtotal of OMB	*12*	*91*	*6*	*13*	*12*	*1*	*17*	*17*	*53*		*3*	*1*	*31*	*3*	*2*	*1*	*2*	*2*	*2*	*2*	*2*	*4*	*2*
PHM	Alternative									15														
	Chilling day									2														
	Chilling hours									6														
	Chilling unit				1																			
	Double-threshold temperature sum							1																
	Forc PO				1					1														
	Forcing and chilling units				1					6														
	Forcing Sarvas									12														
	Forcing thermal time									12														
	°GDD				1			5	1	21														
	°GDH							2		2														
	Heat units									2														
	Parallel									15														
	Parallel Sarvas									11														
	Parallel thermal time									11														
	Q10									1														
	Sequential									15														
	Sequential Sarvas									11														
	Sequential thermal time									11														

	Spring warming											15																	
	State of forcing function				1							1																	
	Temperature sums								1			7																	
	Thermal time											1																	
	Unified forcing											1																	
	Subtotal of PHM				*5*				*9*	*1*	*179*																		
OTM	Bioclimatic index								1		1																		
	Daily mean-based forecast		2								2			2															
	Kriging											6																	
	Kriging + Lr + Sa											1																	
	Land used regression																									3			
	Longitudinal											5																	
	Near neighbor		1																										
	Taxonomic index										1																		
	The nearest monitor method											5																	
	Subtotal of other		*3*						*1*		*4*	*17*	*2*													*3*			
Total		*12*	*94*	*6*	*18*	*12*	*1*		*27*	*18*	*236*	*17*	*5*	*1*	*31*	*3*	*2*	*1*	*2*	*2*	*2*	*2*	*5*		*4*	*2*			

[a] Does not report units of measure. OBM: Observation Based Models. PHM: Phenological Based Models. OTM: other models.

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